F24-017-D-BotForge

Project Team

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Session 2021-2025

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October, 2024

Contents

Chapter 1

Introduction

The main goal of this project is that we are creating a web application that will allow users to create and customize chatbots. Our goal for this document is to present a detailed overview of our project, including its problem statement and our solution, our motivation for this project, its key features, the target audience it is focusing on, and its implementation strategy with a planned timeline. The main objective of this document is to present our vision and perception for BotForge and explain how our project will address and fill the gaps in customer support. It will eventually be very useful for business and organizations by providing them with advanced chatbot technology.

Purpose of the Project:

The main objective for our web application is that we want to create a platform that will help the organizations, businesses and institutions to create and customize chatbots according to their needs on their own with or without any technical knowledge and experience. Our platform will provide an interactive interface that will help and allow its users to upload their data without any difficulty. It will also allow its users to customize the behavior and the appearance of their chatbot they are creating. BotForge will also produce the integration links for the chatbots that users created which will help them in their deployment of their chatbots into their platforms. As there are many complexities in creating chatbots, we will provide an easy path for people to make and customize intelligent and responsive chatbots.

1.1 Problem Statement

As we know that many businesses and organizations face difficulty in providing timely and effective customer support services to their customers. It causes poor experience for customers for these organizations. Since everything and everyone is now opting for AI driven platforms and there is a rise in AI driven platforms, chatbots are proving to be a very important and effective tool for providing good customer services instead of humans. Chatbots provide immediate and compatible help to users but making them and deploying them by integrating them into the platforms tends to be a very costly and difficult task as they require technical knowledge with financial resources. So, the small businesses and startups face issues in providing timely customer support to their users. Moreover, the organizations also face difficulties in making chatbots that align with their needs such as a chatbot that considers and voice their brand. This gap in chatbot development makes it necessary to create a platform that will cater these needs and make the chatbot technology approach to the wider audience.

1.2 Scope

So the main goal that we are trying to achieve from this project is to build a web based platform that is helpful to non technician users and is more beginner-friendly. This project will help to deliver good customer support and help in businesses by providing a interface where users can upload data and build custom chatbots that are trained on the provided data.. So in order to achieve our primary goal we are not focused on provided complex features like AI-driven analytics or multilingual support.

The main features are:

1. **Data Uploading**: Concerning this we will create a space where users will upload data in the form of documents and text cells.
2. **Data Processing**: once the data is uploaded, the platform uses that data to train the chatbots based on the provided data for the purpose of the application.
3. **Chatbot Customization**: So as to spice it up to be more fun and user friendly, the platform will present templates, colors and tones to the users for personalization of

the chatbot design and character.

4. **Integration Link Generation**: During the last stage of the platform, a link will be generated and given to users for use on their websites and other platforms.

1.3 Modules

Our primary objective is to provide a web application that simplifies the process of creating and managing individual chatbots for users. We are breaking up our project into the following components:

1.3.1 Module 1 Data Management Module

This module handles the input and processing of user data, making it accessible and usable for chatbot training and response generation.

- **Feature 1: Data Uploading**
  This feature will help the users so that they can upload their data. data can be product information, FAQs or relevant data. And the major 2 file formats with which users can upload data is, PDF and Text files.
- **Feature 2: Automated Data Cleaning**
  This feature will take the data or the information uploaded by the user and convert it into more clean and preprocessed information so that chatbots get more relevant data.
- **Feature 3: Vector Embeddings Generation**
  Converts the cleaned data into vector embeddings, which are compact numerical representations that the chatbot uses to understand and generate responses.

1.3.2 Module 2 Chatbot Customization Module

This module allows users to personalize the chatbot's behavior, appearance, and response style according to their specific needs.

- **Feature 1: Chatbot Themes**
  Provides a variety of pre-designed templates for chatbot appearances, including different themes, layouts, and color schemes.
- **Feature 2: Styling and Customizing**
  Offers options to select fonts, text sizes, button styles, and other visual elements so that users get the chatbots they prefer and like.
- **Feature 3: Chatbot Personality building**
  Allows users to define the chatbot's conversational tone, such as formal, casual, friendly, or humorous
- This feature helps users so that they can set the tone of that chatbot. Is it formal, casual or more friendly? Users can decide and select it easily.

1.3.3 Module 3 Deployment & Integration Module

This module is about the process of connecting the chatbot over several areas and be ensured that users of the application can access the chatbots easily

- **Feature 1: Live Preview and Testing**
  Provides a live preview and testing environment for users to see how the chatbot will function before final deployment.
- **Feature 2: Generation of Chatbot Links**
  This feature will create a link that will connect the users with the chatbot. It means this link can be connected to any website or application and will help to access the chatbot through this link

1.3.4 Module 4 Payment Methods

This module focuses on preparing the chatbot for deployment, testing it for user feedback, and ensuring its integration into websites and platforms.

- **Feature 1: Secure Payment Processing**
  Implements robust security measures such as encryption, tokenization, or compliance

1.4 User Classes and Characteristics

User Class	Characteristics
Small Business Owners / Entrepreneurs	• Limited technical expertise. • Interested in cost-effective, easy-to-deploy customer support solutions. • Need to customize chatbots to reflect their brand's tone and style.
Customer Support Managers	• Moderate understanding of technology but may not have programming skills. • Looking for efficient ways to handle frequent customer queries. • Interested in chatbots for automating FAQs
Non-Technical Users	• Minimal to no coding experience. • Need a simple and intuitive interface to create and deploy chatbots without technical knowledge. • Interested in automating basic customer interactions
Educational Institutions	• Use chatbots to provide automated support for students and staff • May have limited technical resources and look for user-friendly solutions.
Technical Experts	• Familiar with the underlying technology of chatbots and may want more control over customization. • To speed up the development process by utilizing pre-built functionalities.

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Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case/Event Response Table/Storyboarding

The Use-cases of Bot-Forge are given as follows:

2.1.1 Use Cases

1. **Register and Login Account**

Use Case Name	Register and Login Account
Actors:	User, System, Database
Type	Primary
Description:	Users create an account or log in using their credentials. The system ensures secure authentication for accessing the platform.

2. **Handle User Authentication**

Use Case Name	Handle User Authentication
Actors:	System, Database
Type	Supporting
Description:	The system verifies user credentials during login and registration. It interacts with the database to ensure secure authentication and access

	control.

3. Manage User Accounts

Use Case Name	Manage User Accounts
Actors:	System, Database
Type	Supporting
Description:	The system manages user accounts, allowing users to update personal information, view payment history, and manage their chatbot projects. It interacts with the database to retrieve and update user details.

4. Upload Data

Use Case Name	Upload Data
Actors:	User, System, Database
Type	Primary
Description:	Users upload files (PDFs, text documents) containing data that will be used to train the chatbot. The system validates the file format and ensures successful upload for further processing.

5. Store User Data

Use Case Name	Store User Data
Actors:	System, Database
Type	Supporting
Description:	The system stores user-related data, such as uploaded files, chatbot configurations, and credentials, in the database for future retrieval.

6. Preprocess Data

Use Case Name	Preprocess Data
Actors:	System

Type	Supporting
Description:	The system preprocesses user-uploaded data, cleaning it and preparing it for further use in chatbot training. This involves removing noise, formatting inconsistencies, and tokenization.

7. **Select Chatbot Template**

Use Case Name	Select Chatbot Template
Actors:	User, System
Type	Primary
Description:	Users choose from pre-designed chatbot templates to define the appearance and layout of their chatbot. The system ensures the selected template is applied correctly.

8. **Customize Chatbot Appearance**

Use Case Name	Customize Chatbot Appearance
Actors:	User, System
Type	Primary
Description:	Users personalize the visual aspects of the chatbot, such as colors, fonts, and button styles. The system implements these customizations into the chatbot interface.

9. **Select Chatbot Tone**

Use Case Name	Select Chatbot Tone
Actors:	User, System
Type	Primary
Description:	Users select the conversational tone for their chatbot (e.g., formal, casual, friendly). The system applies this tone to the chatbot's response generation.

10. **Generate Vector Embeddings**

Use Case Name	Generate Embedding Generation
Actors:	System
Type	Supporting
Description:	The system generates vector embeddings from cleaned user data. These embeddings are numerical representations that help the chatbot understand and respond to queries.

11. Store Vector Embeddings

Use Case Name	Store Vector Embeddings
Actors:	System, Database
Type	Supporting
Description:	The system generates vector embeddings from user-uploaded data and stores them in the database. These embeddings are later used for chatbot response generation.

12. Retrieve Chatbot Data

Use Case Name	Retrieve Chatbot Data
Actors:	System, Database
Type	Supporting
Description:	The system retrieves previously stored chatbot data, such as configuration settings, vector embeddings, and user preferences, from the database for further customization or deployment.

13. Preview Chatbot

Use Case Name	Preview Chatbot
Actors:	User
Type	Primary
Description:	Users preview the chatbot in a simulated environment to test its appearance and behavior before deployment. The system generates a real-time preview for user validation.

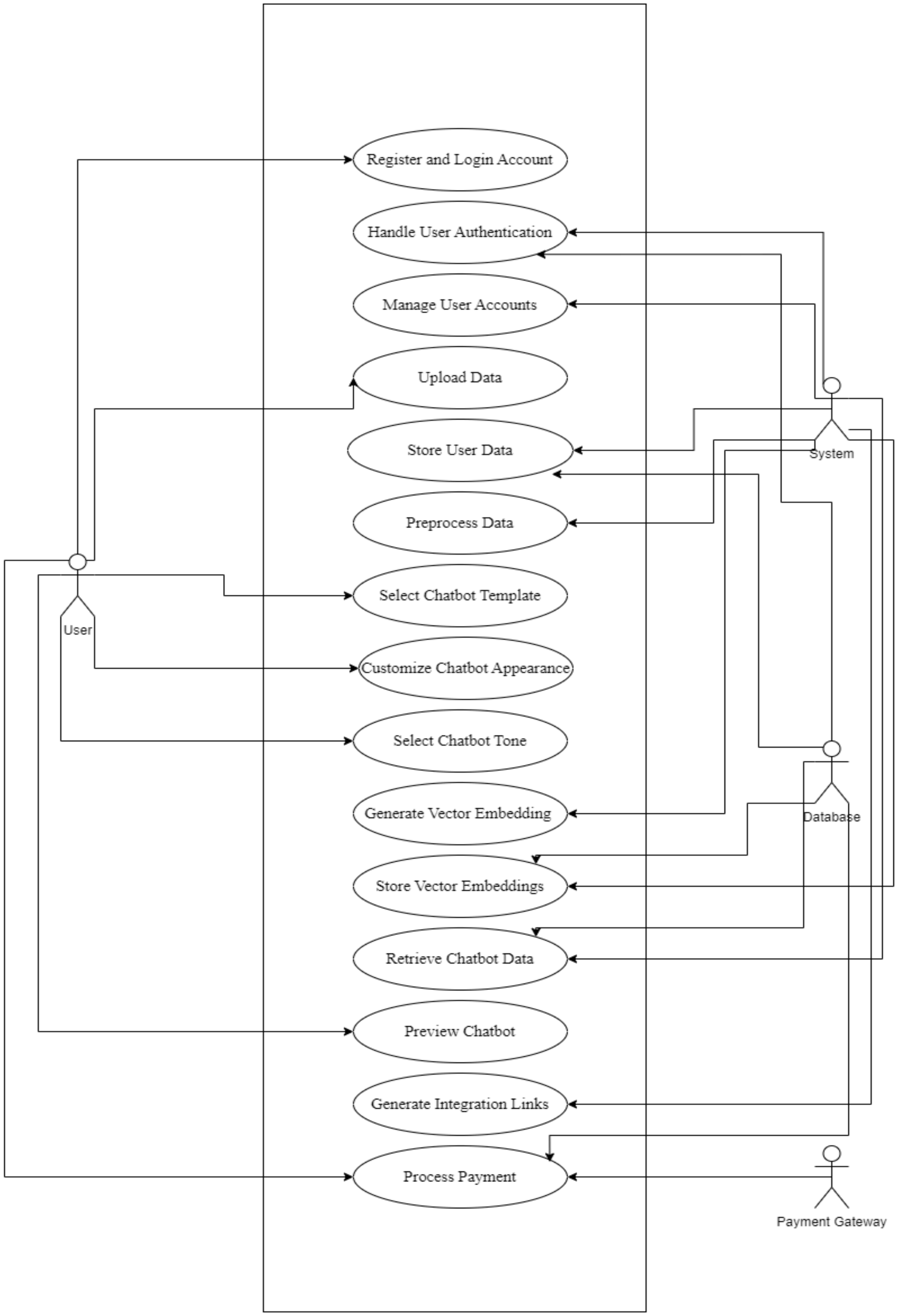
14. Generate Integration Links

Use Case Name	Generate Integration Links
Actors:	System
Type	Supporting
Description:	The system generates a unique integration link for each chatbot. Users can embed this link into their websites or applications to deploy the chatbot.

15. Process Payment

Use Case Name	Process Payment
Actors:	User, Payment Gateway System
Type	Primary
Description:	Users process payments for premium features or subscriptions. The system interacts with the payment gateway to securely process the transaction and stores payment details in the database.

2.1.2 Use Case Diagram



2.1.3 Expanded Use Cases

1. **Register and Login Account**

<table>
<tr><td>Use Case Name</td><td>Register and Login Account</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Actors</td><td>User, System</td></tr>
<tr><td>Stakeholders and Interests</td><td>User: Wants to create an account or log into an existing one to access the platform's features.
System: Needs to securely create user accounts and ensure secure authentication.</td></tr>
<tr><td>Pre-conditions</td><td>The registration or login page is accessible.
The system is operational, and email verification services are working (for registration).</td></tr>
<tr><td>Post-conditions</td><td>A new account is successfully created, and the user receives a verification email.
The user is authenticated and granted access to their account.</td></tr>
<tr><td>Main success scenario</td><td><table>
<tr><th>User</th><th>System</th></tr>
<tr><td>The user enters their details (email, password, and other required information) on the registration page.</td><td></td></tr>
<tr><td></td><td>The system checks for duplicate email addresses.</td></tr>
<tr><td></td><td>If valid, the system creates the account and sends a verification email.</td></tr>
<tr><td>The user clicks the verification link in the email to activate their account.</td><td></td></tr>
</table></td></tr>
</table>

	<table><tr><td></td><td></td></tr><tr><td></td><td>The system authenticates the credentials and logs the user into the platform.</td></tr></table>
Alternatives	• If the email is already registered, the system shows an error and prompts the user to log in or use a different email. • If the account is not yet verified, the system prompts the user to verify their email before logging in.
Technology and Data Variation Lists	Email and password.
Frequency of Occurrence	Users register once but may log in frequently as they access the platform.

2. **Handle User Authentication**

Use Case Name	Handle User Authentication
Level	Sub-function
Actors	System, User
Stakeholders and Interests	**User**: Wants to log in securely to access their account and chatbot-related data. **System**: Needs to ensure that the user is authenticated and authorized securely to access the platform's resources.
Pre-conditions	• Users have an account registered on the platform. • System is ready and operational, and the login page is accessible.
Post-conditions	• User is authenticated and granted access to their account. • Authentication token/session is created for the user session.
Main success scenario	

<table>
<tr><td></td><td>
<table>
<tr><th>User</th><th>System</th></tr>
<tr><td>User enters their login credentials (username/email and password) on the login page.</td><td></td></tr>
<tr><td></td><td>System checks the credentials against the stored records in the database.</td></tr>
<tr><td></td><td>If the credentials are valid, the system authenticates the user and creates a session.</td></tr>
<tr><td>The user is redirected to the dashboard or homepage.</td><td></td></tr>
</table>
</td></tr>
<tr><td>Alternatives</td><td>If the password entered is incorrect, the system displays an error message and allows the user to try again.
If the user has forgotten their password, they can request a password reset via email.</td></tr>
<tr><td>Technology and Data Variation Lists</td><td>Username & password combination, multi-factor authentication</td></tr>
<tr><td>Frequency of Occurrence</td><td>Every time the user attempts to log in or during session expiration.</td></tr>
</table>

3. **Upload Data**

Use Case Name	Upload Data
Level	User Goal
Actors	User, System

<table>
<tr><td>Stakeholders and Interests</td><td>User: Wants a quick and easy way to upload data for training the chatbot.
System: Needs to ensure that the uploaded data is correctly processed, stored, and prepared for chatbot training.</td></tr>
<tr><td>Pre-conditions</td><td>User is logged into their account.
The system is operational, and the data upload interface is accessible.
The user has a file in the correct format (PDF or Text) ready for upload.</td></tr>
<tr><td>Post-conditions</td><td>The data is successfully uploaded, processed, and stored in the system for chatbot training.
User receives a confirmation of the successful upload.</td></tr>
<tr><td>Main success scenario</td><td><table>
<tr><th>User</th><th>System</th></tr>
<tr><td>The user accesses the “Upload Data” section of the platform.</td><td></td></tr>
<tr><td>The user selects the file they wish to upload (PDF or Text).</td><td></td></tr>
<tr><td></td><td>The system validates the file format and checks for any issues.</td></tr>
<tr><td></td><td>The system uploads and stores the data.</td></tr>
<tr><td>A confirmation message is shown to the user, indicating the success of the upload.</td><td></td></tr>
</table></td></tr>
</table>

Alternatives	If the file is in an unsupported format, the system shows an error message and prompts the user to upload a valid file.
Technology and Data Variation Lists	Data can be in PDF or text formats. Users may upload from desktop or mobile devices.
Frequency of Occurrence	Everytime a new chatbot is created

2.2 Functional Requirements

The functional requirements of BotForge are as follows:

2.2.1 Data Management

This module is responsible for handling user data, ensuring it is uploaded, processed, and stored for chatbot training.

1. **FR1: Data Uploading**
   Description:
   The system shall allow users to upload data in supported formats (PDF, Text).
   Details:
   - Users can upload data files (product information, FAQs) that the chatbot will use for training.
   - The system shall validate the file format and file size before processing.
   - The system shall provide feedback in case of unsupported file formats or size limits.

2. **FR2: Automated Data Preprocessing**
   Description:
   The system will perform data preprocessing to the data uploaded by the user and make it ready for the training of the chatbot.
   Details:
   - The data will perform filtering, cleaning, and formatting to eliminate extra elements like stop words and HTML tags.
   - Users will be informed by the system when preprocessing is successful or when errors occur.

3. **FR3: Data Storage**
   Description:
   The processed data will be then stored in the system for future use and references.
   Details:
   - Data will be kept in a format (like vector embeddings) that can be used to train chatbot models.
   - When user data is stored in a database, the system must make sure it is safely encrypted.

4. **FR4: Vector Embeddings Generation**
   Description:
   The system will create chatbots by converting the processed data in vector embeddings.
   Details:
   - The system will create embeddings that reflect the meaning of user data by using the vector data.
   - When responding to chatbots, the embeddings must be kept in a format that is optimized for quick responses.

2.2.2 Chatbot Customization

This module is focused on allowing users to personalize the appearance, behavior, and tone of their chatbots.

1. **FR1: Chatbot Theme Selection**
   Description:
   Pre-designed themes will be provided for chatbot outlook by the system.
   Details:
   - A template can be selected for the chatbot that will suit users interest.
   - They can also check their theme before selecting it.

2. **FR2: Appearance Customization**
   Description:
   The users can enhance the chatbot outlook by customizing it.
   Details:
   - The font,colors,text sizes and buttons will be provided for customization.
   - Users can change and set the appearance of the chat window and the chatbot icon.

3. **FR3:Chatbot Tone Selection**
   Description:
   The system shall allow users to customize the tone and method of the chatbot's responses.
   Details:
   - Users can choose a tone which can be casual or formal and depends on the user's needs to reflect their brand or audience preferences.
   - The chatbot can be customized to be more empathetic, professional, or friendly based on the user's preference.
   - The system shall show how the chatbot responds and answers with the selected tone before deployment.

2.2.3 Deployment and Integration

This module is for the final deployment of chatbots and making their integration links for different platforms.

1. **FR1: Integration Link Generation**
   Description:
   Integration links will be created for chatbots so they can be put into the respective platform of users.
   Details:
   - A special URL or API endpoint will be provided to users so they can integrate the chatbot on their platforms.
   - The system is responsible for making sure the integration link functions

flawlessly with the application or website where it is installed.

2. **FR2: Live Chatbot Preview and Testing**
   Description:
   The system shall provide a live preview feature for users to test their chatbots before deployment.
   Details:
   - Users can interact with the chatbot in real-time to check its behavior, tone, and responses.
   - The system shall allow users to make adjustments based on the preview.

2.2.4 Payment Methods

This module will perform the payment techniques and handle user subscriptions.

1. **FR1: Payment Processing**
   Description:
   Payments will be made for some features and they will be secure.
   Details:
   - Users will enter their payment details and those payments will be processed by a payment gateway system and the information will be encrypted.

2. **FR2:Subscription Management**
   Description:
   The system will allow the users to select and manage their subscriptions.
   Details:
   - Users can change their plan of subscription based on their use.
   - The system shall notify users of upcoming payment dates and any issues with their payment methods.

2.2.5 User Account Management

This module manages user profiles and authentication.

1. **FR1: Register and Login Account**
   Description:
   The system shall allow users to register and log in securely.
   Details:
   - Users can create accounts using their email and password.
   - The system shall validate the login credentials and authenticate users securely.
   - The system shall provide a password reset option for users who have forgotten their credentials.

2. **FR2:Handle User Authentication**
   Description:
   The system shall handle user authentication and session management.
   Details:
   - The system shall ensure secure user authentication using encrypted credentials.
   - Session management features will log users out after a period of inactivity.

3. **FR3:Manage User Profiles**
   Description:
   The system shall allow users to manage their personal information and account settings..
   Details:
   - Users can update their account details (email, password, etc).
   - The system shall notify users of any changes made to their profiles.

2.3 Non-Functional Requirements

Following are the requirements that fall under the Non functional terms. So in order to maintain our system "Botforge" as a standard system that provides efficiency in terms of security, performance and usability we have ensured the following non functional requirements for our system. Each requirement will help to enhance the overall system working.

2.3.1 Scalability Requirements:

Horizontal Scaling: Our system database and server should be designed in such a way that it would be capable of adding more and more servers or resources being added into it. So in accordance with the user demand our system will be more scalable.

Data Storage: As the number of uploaded data increases and users make an increased number of chatbots, our system should have enough storage to store the required data and run efficiently.

2.3.2 Usability Requirements:

Ease of Use: The platform should be very easy to use such that if any person with no background knowledge and technical expertise accesses this system he/she will be able to navigate the platform easily.

UI Responsiveness: The user interface should be responsive, it should cover main possible screen sizes such as desktop, tablet and mobile screen size.

2.3.3 Performance Requirements:

Response Time: The system should be very fast so that when any user requests for chatbot customization and data uploads, all results should be provide to the user within the 20 seconds.

Scalability: The platform should be able to handle multiple side by side requests from at least 15 users who are accessing the platform at a time.

2.3.4 Security Requirements:

Data Encryption: All the data that is to be sensitive, which may include data related to user such as name, email, phone number. Data related to payment methods, like card numbers and pins much be secure with encryption methods related to the system requirements.

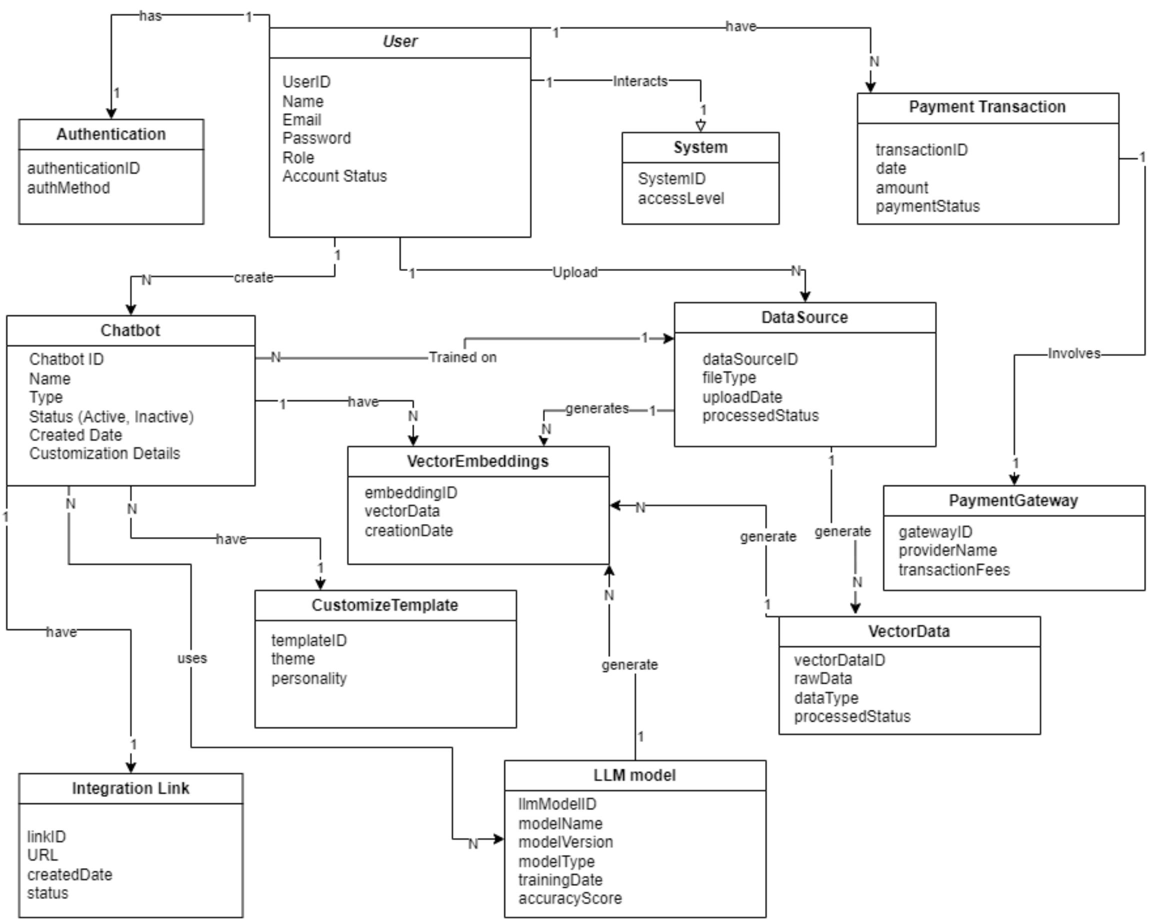
Authentication: Users must provide the information such as email and password before accessing the system so that to keep authorized users away from the platform.

Data Privacy: The platform must guarantee the privacy of the user data.

2.3.5 Integration Requirements:

Third-Party APIs: The system should integrate with third-party APIs, such as payment gateways, in the most smooth way so that users do not face any delay in the requests been processed.

2.4 Domain Model



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Chapter 3

System Overview

In the System Overview section, it gives the high-level description, which covers the general purpose, key functionalities, and design. In this part, how the system is going to operate, the problems it is going to solve, and for whom it will work are described. The core system components are also mentioned along with their respective interactions, introducing the architecture as a whole. The background information relevant to understanding the objectives of a project, or more to the point, the technology involved, comes in for outlining the scope and importance of the system.

3.1 Architectural Design

For our project **Botforge** we would be using **Multi-tiered architecture** as our system includes some separate layers all connected together and having a specific purpose. i.e. data handling, chatbot customization, and deployment.

Here is how we have structured our Architecture:

There will be mainly 4 layers involved in our system,

- Presentation Layer
- Application Layer
- Data Management Layer
- Integration Layer

Here is the breakdown of each layer showing the main purpose and how each layer contributes to the overall system working and performance.

Presentation Layer:

Components: Web UI, Chatbot Customization Interface, Data Upload Interface.

This is the end-user facing side where user can actually look and feel the interface, perform necessary action required i.e. uploading data, customizing chatbot etc

Application Layer:

Components: Data Processing (Data Preprocessing, Embedding Generation), Chatbot Generation, Customization Engine.

This layer consists of the core functionality of the overall processing and backend functionality of how uploaded data is processed, models are trained, chatbots appearance and customization of behavior works.

Data Management Layer:

Components: Embedding Store, Vector Database, File Handling.

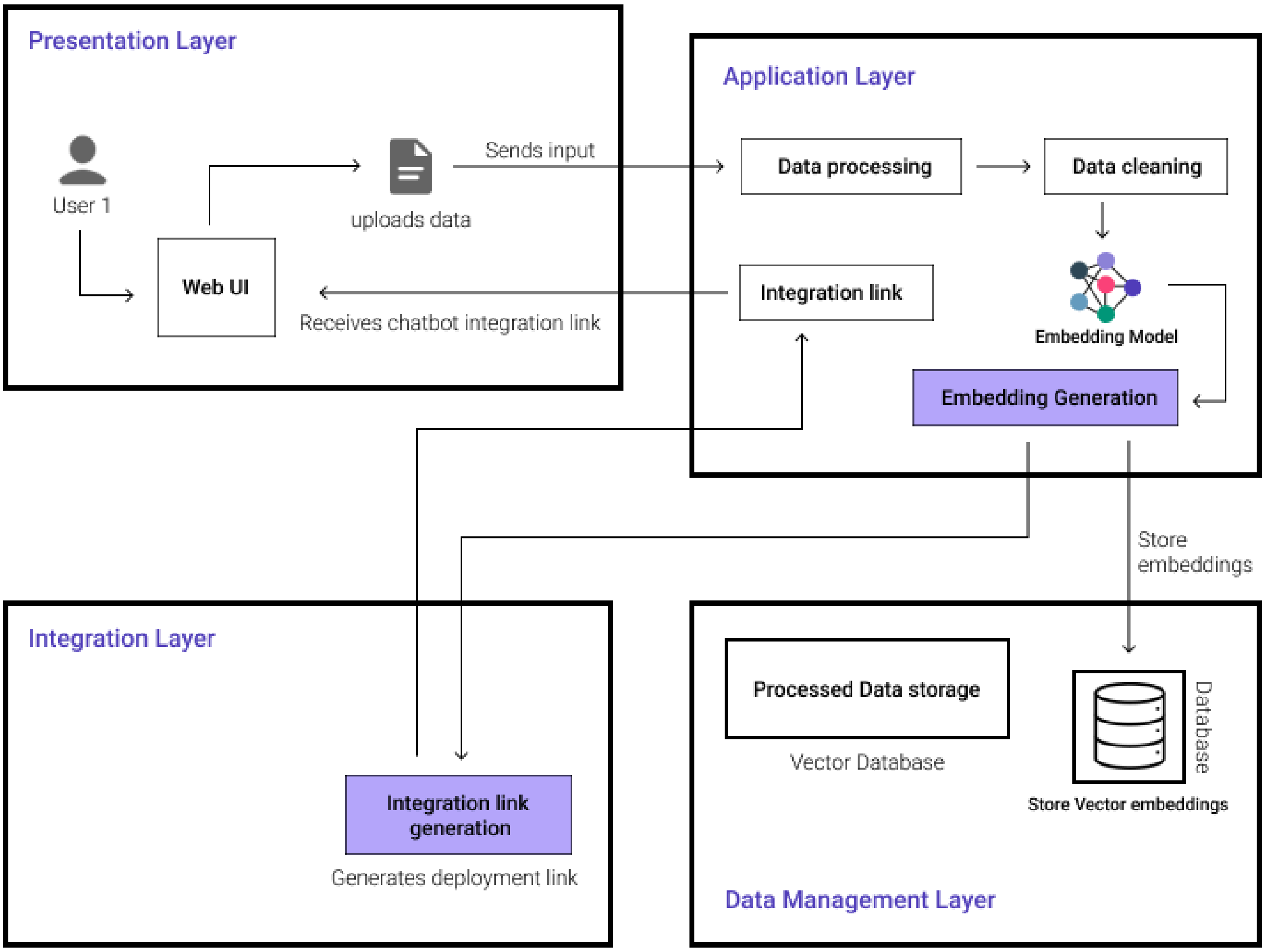
This layer handles how processed data is stored and retrieved. The embeddings and vectorized data generation from users uploaded data is stored here.

Integration Layer:

Components: Chatbot Link Generator, APIs.

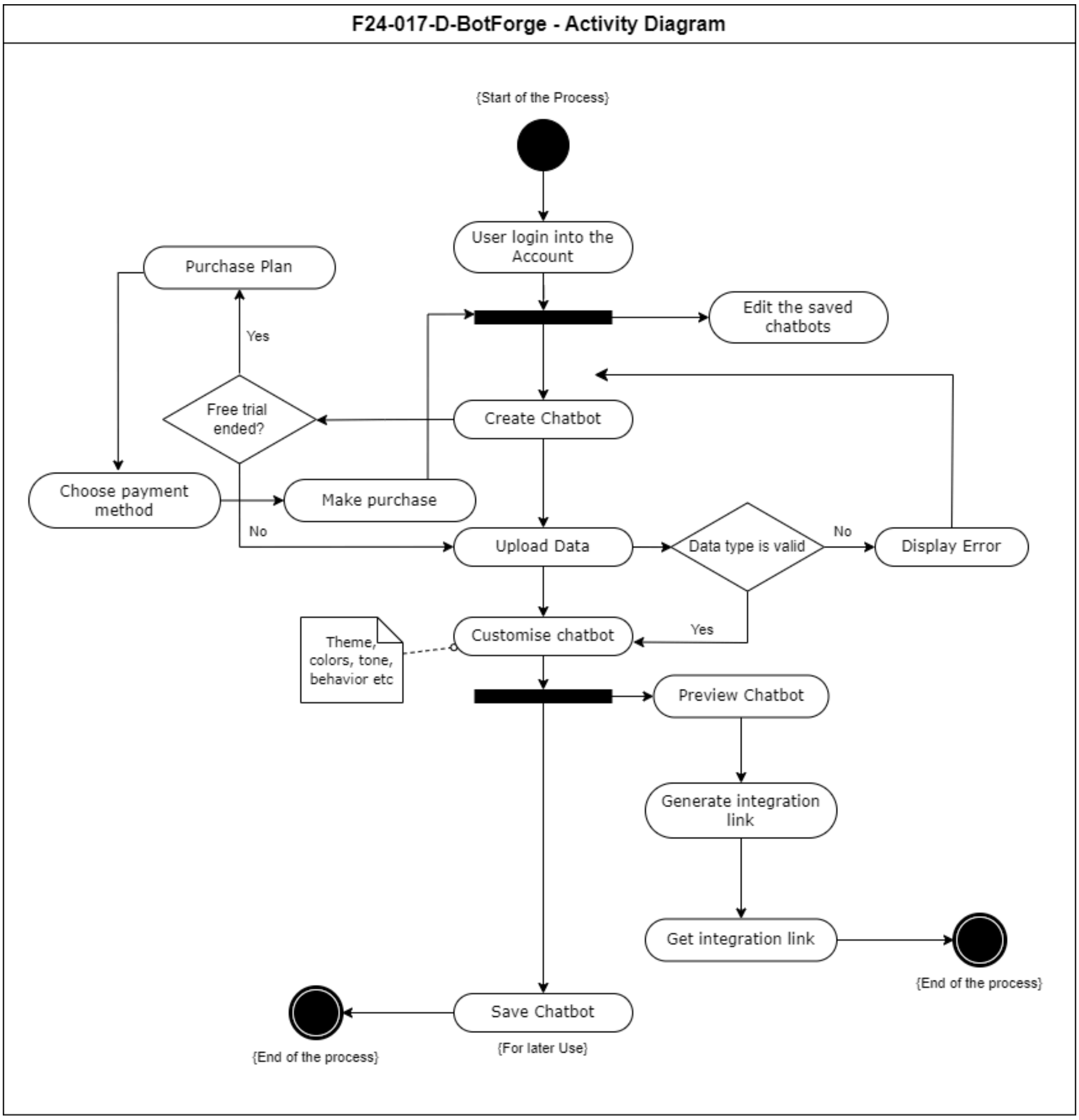
This layer handles the integration process allowing the generation of embedded links for the chatbots created.

Diagrammatic Representation:



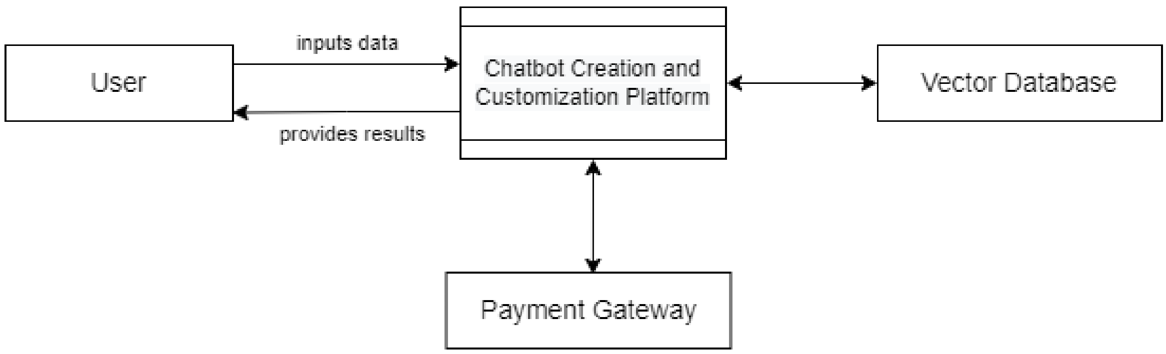
3.2 Design Models

Activity Diagram:



Data Flow Diagram:

Level 0 Diagram of F24-017-D-BotForge
Context Diagram



Here the Context diagram shows the high-level overview of the system and actual representation of the whole system as one process and display the external entities as the actors interacting with the system to show the main purpose of the system.

External Entities:

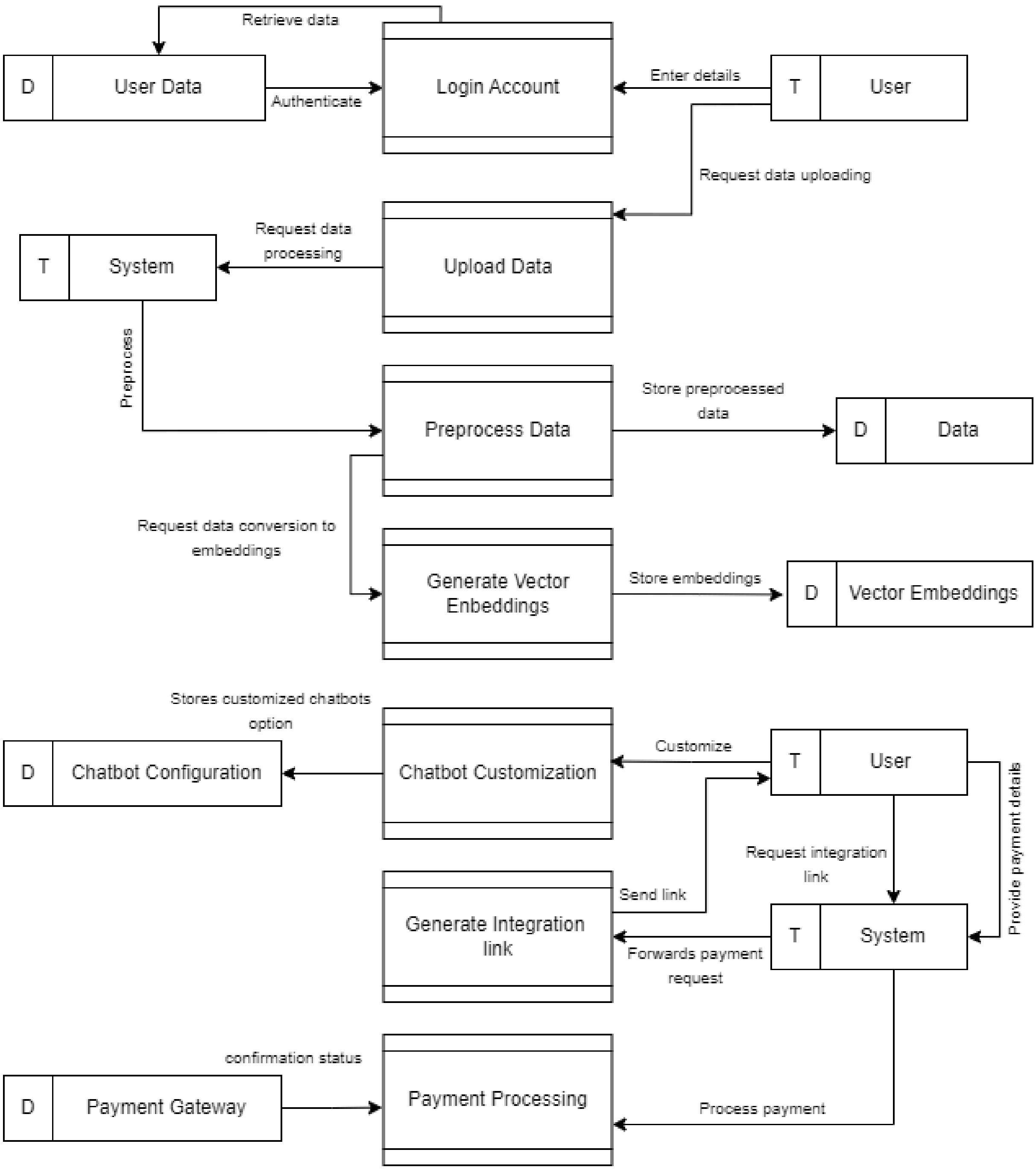
User: The actual person who is going to interact with the system in order to build and customize the appearance of their chatbot and generate integration links.

Payment gateway: This entity is going to process payments for the users to access and purchase plans for more than a limited number of chatbot creation.

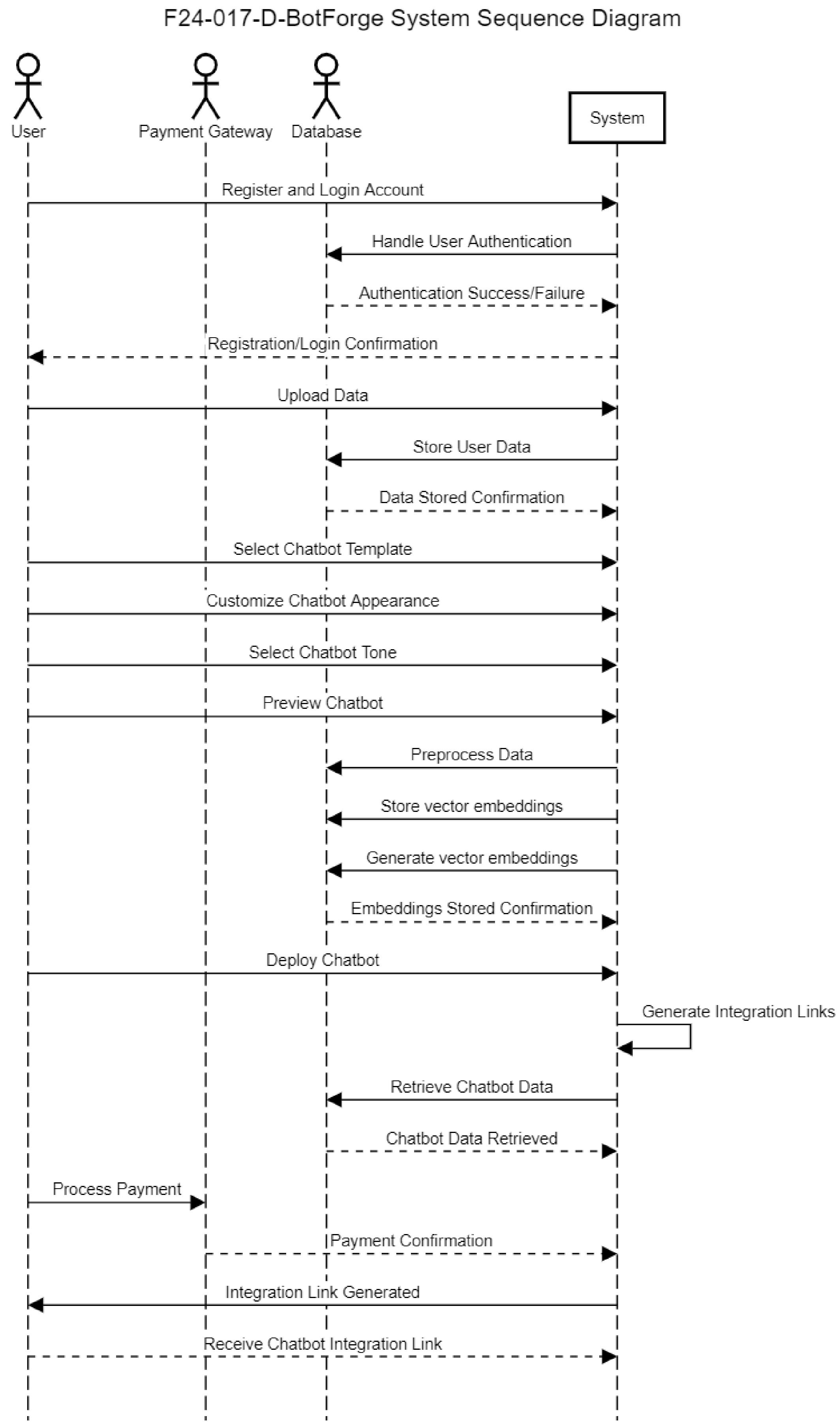
Vector database: Stores and helps in retrieving vector embeddings and preprocessed data.

Level 1 Diagram of F24-017-D-BotForge

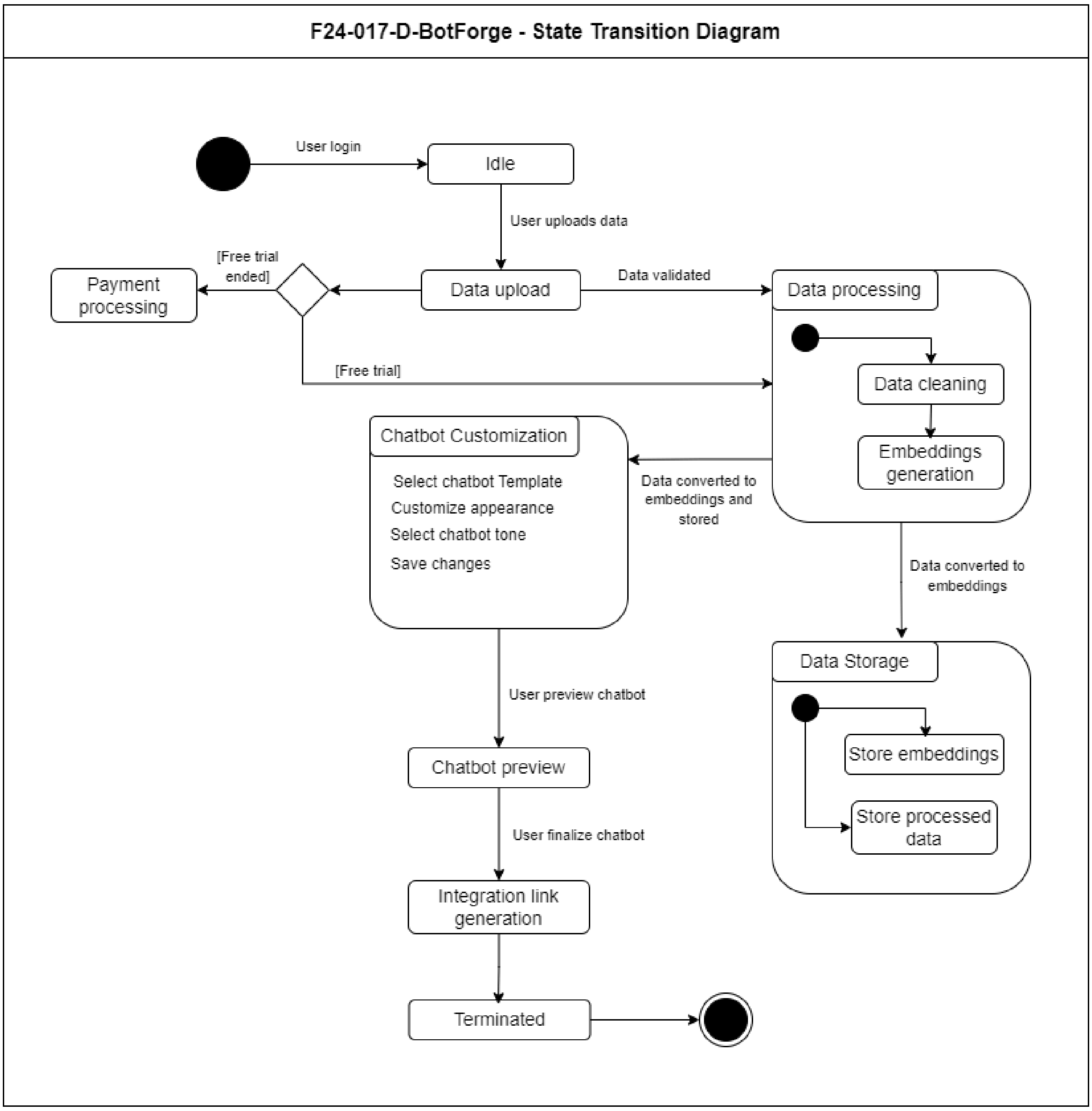
D [DataBase] ⟷ [All processes]



System-level Sequence Diagram:



State Transition Diagram



3.3 Data Design

For our project **Botforge**, We have to understand how the data being involved in making of the whole system is converted into organized data structures that are simple to store, process, and retrieve in order to set up our BotForge project.

Overview of Information Domain:

The Data stored into the information domain of our project Botforge includes:

User Information: This is the information relate to the User such as user information that is required to be stored in order to create user account and manage payment details. The number of chatbots the user has made may also be considered in it.

Chatbot Data: This data is all about the chatbot itself. What is the domain of the chatbot what data is used to train the chatbot, chatbots integration link all are considered data related to chatbots.

Uploaded Data: This is the information uploaded by the user in order to build chatbots. this is the data such as FAQs, product information or relevant data used to train chatbots.

Payment Information: Information related to payment processing, user account information and gateways data is included.

Transforming into Data Structures:

This above mentionable data and the information to be stored into database needs to be given a proper format and organized format so we have converted it into the structured way, here is how:

Data Modeling: We have used the concept of data modelling this represents using data models that can easily define how two or more then two entities are related to each other and what sis the relationship between them. For example if we have one user model so it is related to chatbot model as: One user can create multiple chatbots,

Data Structures:

User Structure: This includes elements and attributes like userID, username, password, and preferences.

Chatbot Structure: This includes chatbotID, userID, templateID, customizations, and trainingData.

Uploaded Data Structure: This includes dataID, userID, filePath, and dataType.

Payment Structure: This includes paymentID, userID, amount, status, and transactionDate.

Storage and Organization:

Once the data domain is transformed into data structures i.e presented in the well-formatted way, it can now be stored into database for easy and timely fast access for the user using the system, in order to satisfy the non functional requirements of the system mentioned in section 2.3.

Databases:

User Database: Here all the data related to the user using and interacting with the system is stored. We are using SQL database for this purpose.

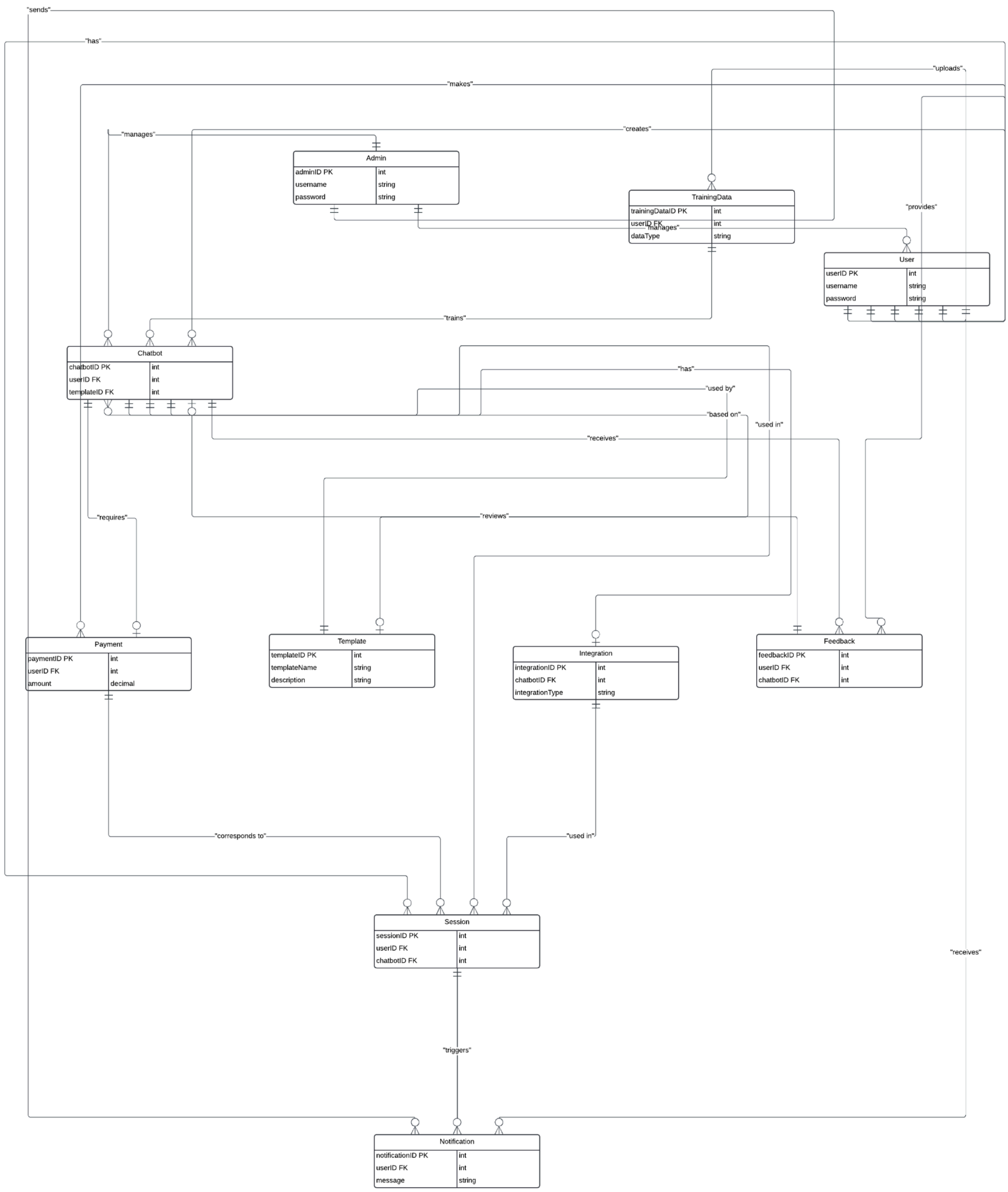
Chatbot Database: This database section stores the information related to the chatbots that help to identify each chatbot uniquely so that the chatbots can easily be modified and retrieved form the database easily and can be used by users in the future.

Data Upload Storage: This can be cloud storage or a file system where the uploaded documents are stored this also incudes the preprocessed data after the successful processing and embeddings generation by the system.

Vector Database: There is the need of storing vector embeddings generated after the preprocessing of the uploaded data. This is stored in the vector database.

Payment Database: Keeps track of all payment transactions and status.

Entity-Relationship Diagram:



Link to ERD Diagram:

F24-017-D-BotForge_ERD_Diagram

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Author: Various Contributors
Journal: Software Engineering Stack Exchange
Year: 2011
URL: https://softwareengineering.stackexchange.com/questions/62965/what-is-the-definition-of-user-classes-with-respect-to-software-use
Owner: Not found

Title: What is DFD(Data Flow Diagram)?
Author: GeeksforGeeks
Journal: GeeksforGeeks
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URL: https://www.geeksforgeeks.org/what-is-dfddata-flow-diagram/

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